Sammy Mustafa

Effect of Immune Checkpoint Blockade on Adaptive Immunity in Renal Cell Carcinoma

Introduction

Renal cell carcinoma (RCC) is the most common type of kidney cancer in adults[1] and is highly infiltrated with immune cells like T-cells.[2] Thus, recent research has focused on studying the composition of the tumor microenvironment and its response to current first-line treatments like immune checkpoint blockade (ICB).[3] Specifically, Bi et al. have demonstrated how ICB remodels the RCC microenvironment in order to better understand the effects of ICB exposure and responses.[4] To better understand the study's findings surrounding the effect on immunity from ICB treatment with several responses, the single-cell RNA sequencing (scRNA-seq) methodologies leveraged will be reviewed and attempted to be replicated.

The tissue source for the study was tumor tissue from patients with RCC. Samples were collected with informed consent and ethics approval, and included information on patient metadata such as age, sex, treatment regimen, ICB/TKI exposure, clinical response, and more. The experimental approach involved mechanical and enzymatic dissociation for scRNA-seq. Specifically, solid tumor tissue was collected and an enzymatic dissociation mix (with Collagenase Type 4 and DNAse I) was used. The tissue was minced and incubated with rotation, followed by red blood cell lysis, viability assessment, and cell loading onto the 10x Genomics Chromium platform using Single Cell 3' reagents. The Illumina's Rapid Capture Exome Kit was used for whole exome sequencing of some samples. Alignment was done using BWA software, somatic mutation calling with MuTect, and copy number alteration calling using FACETS. The scRNA-seq samples were sequenced on an Illumina HiSeq X sequencer, with demultiplexing, barcode processing, read alignment, and UMI counting done using the 10x Cell Ranger analysis pipeline. After initial processing, droplets suspected of containing more than one cell were identified and excluded using the Scrublet tool, ensuring the removal of potential doublets which could skew the data interpretation. Subsequently, an ambient RNA decontamination step was performed using the SoupX package to precisely subtract ambient RNA profiles, based on specific gene markers, from each cell's data to enhance the accuracy of the single-cell RNA expression profiles.

Methodology

After the paper's quality control, there were 34,326 total cells. However, the RNA count and feature data were visualized (*Figure 1*) and several extreme outliers were identified. Thus, the data was subsetted to include 500-19,000 RNA counts and 200-4000 features; this specification was selected as it kept 32,874 cells and did not drastically lower the cell count. To overview the entire dataset, clustering was done on all cell types to understand how to handle the data. Following normalization, finding the genes that differ the most across cells, scaling to make the genes more comparable, dimensionality reduction using PCA, and identifying the suitable number of principal components from an Elbow Plot for the UMAP (30), the metadata revealed the potential batch effects from the different patients and organs (as they are both independent of the analysis) the data were extracted from; with this, harmony was run on these parameters for batch correction. From this, a resolution was identified to cluster the data based on the 33 clusters identified by Bi et al. (*Figure 2*) and the top marker genes were identified for each cluster (*Figure 3*). However, considering our focus is on the effects of ICB treatment on immunity, we needed to stratify the data by cell type (*Figure 4*). After identifying the cells involved in both adaptive (lymphoid) and innate (myeloid) immunity, the data was subsetted by these two cell types for our analyses.

The pipeline of finding variable features, scaling, dimensionality reduction with PCA/UMAP, and batch correction using harmony was then repeated on these two subsets to be able to obtain more specific marker genes within the cell types individually for clustering. The resolutions selected for each cell type were selected based on trial-and-error; the resolutions selected have more clusters than the reported paper (15 vs. 13 for lymphoid cells and 14 vs. 9 for myeloid cells) as higher resolutions were needed to isolate certain clusters identified by the paper. The top marker genes of the lymphoid cells (*Table 1*) uncovered 11 clusters defined by different lymphoid cell types (*Figure 5*). These clusters included several different CD8+ T-cell types, NKs, Memory/Effector T-Helpers, plasma cells, and more. The top marker genes of the myeloid cells (*Table 2*) uncovered 8 clusters defined by different myeloid cell types (*Figure 6*). These clusters included CD16 monocytes, dendritic cells, tumor-associated macrophages, and more. The top marker genes from both cell types were linked to cell types/annotated using marker gene databases,[5] matched with the cell-type-defining genes from Bi et. al.,[4] and further confirmed with additional scientific studies. It is important to note our higher resolutions broke up well-defined clusters which were combined during the annotation process; they also did not recognize every cell type defined by the paper, potentially due to our QC or errors in the original paper.

Downstream Analyses

I was then interested in the differences surrounding adaptive immunity from ICB exposure as well as from different responses to ICB treatment. Thus, I sought out to do a differential expression analysis of different treatment/response groups to answer "How do the most upregulated human hallmark pathways differ from different ICB exposures and responses in lymphoid cells?". Thus, I subsetted the data by lymphoid cells to focus on adaptive immunity and then identified the differentially expressed genes that were more highly expressed from each subtype of 1) ICB exposure (ICB treatment and no ICB treatment) and 2) ICB response (progressive, partial, stable, no treatment, and N/A) using the Wilcoxon Rank Sum test. From this, I compared the human hallmark gene sets to those identified from each subtype to find the most upregulated genes and human hallmark pathways for each subtype (ranked by normalized enrichment score). From this, I compared the top three pathways across each subtype to see what it underscored about each subtype.

The top three highly expressed human hallmark pathways from RCC patients with and without ICB treatment (*Table 3*) differ in terms of what they are representing. The pathways for no ICB treatment (coagulation, angiogenesis, and inflammatory response) align with the uninterrupted growth of tumor cells and response of inflammation without treatment; this is done in a way that gives the tumor a hypercoagulable state that can lead to its progression in blood vessels.[6, 7]

Interestingly, the pathways for ICB treatment (E2F targets, G2M checkpoint, MYC targets) also suggest greater proliferation, migration, and invasion of RCC cells with a stress response.[8, 9] While these pathways may still support a pro-tumorigenic environment, these changes represent a change in the lymphoid cells that may still slow the tumor progression and growth.

The top three highly expressed human hallmark pathways from RCC patients receiving ICB treatment differed based on their response to the treatment (*Table 4*). The partial response to ICB treatment, in which least a 30% decrease in the sum of the longest diameter (LD) of target lesions (taking as reference the baseline sum LD), had highly expressed pathways almost identical to those identified to general ICB treatment, with the interferon alpha response pathway upregulated instead of the MYC targets. This highlights the antiviral, antitumor, and stress responses in this subtype with the least growth due to the active immune damage.[10] The progressive subtype, in which there is at least a 20% increase in the sum of the LD of target lesions, has pathways (myogenesis, epithelial-mesenchymal transition, and TGF-beta) that align with promoting fibrosis and invasiveness of RCC.[11, 12] The stable response to ICB treatment, in which there is neither significant shrinkage or growth, has similar pathways to the progressive response with the addition of the TNFα Signaling via NFKB pathway; in this context, these pathways may suggest growth combating against the tumor suppression from the ICB treatment.[13] Overall, the most highly expressed human hallmark pathways do differ for each ICB treatment and response and align with their corresponding phenotype and the pathways' reported effects on lymphoid cells in RCC.

For my second question, I wanted to identify whether there were significant differences in the proportion of each cell type due to ICB exposure and responses for adaptive immunity. Thus, I sought out to do differential abundance analysis of different treatment/response groups to answer "How does ICB exposure and response affect cell abundance in lymphoid cells? Are there significant differences in cell abundance of specific cell types?". Again, I subsetted the data for lymphoid cells only and got the proportion of each cell type for each subtype of ICB exposure and response. After visualization of the varying cell abundances across each subtype, a single cell type within the lymphoid cell types was used to conduct statistical hypotheses tests comparing the abundance of that cell type's cells between two groups of interest in order to confirm observed visual differences.

The cell abundance by ICB exposure (*Figure 7*) visualizes how ICB treatment has a greater proportion of 41BB-Lo CD8+ T cells compared to the no treatment population. This difference aligns with how tumor-infiltrated CD8+ T cells have a long-term progenitor exhausted population that can differentiate into exhausted cells from immune checkpoint inhibition therapy.[4, 14] To confirm this difference between groups, a Pearson's Chi-squared test was conducted for 41BB-Lo CD8+ T cells (over a Fisher's test considering counts are higher than 5) across the ICB exposures: the χ^2 was 458.1 and the p-value was 2.2e-16. These results reject the null hypothesis that there is no difference in the cell abundance of 41BB-Lo CD8+ T cells across ICB treated and non-treated samples. The extremely low p-value and the very high χ^2 value suggests that ICB treatment can significantly increase the proportion of 41BB-Lo CD8+ T cells among lymphoid cells.

More specifically, the cell abundance by ICB response (*Figure 8*) visualizes how certain ICB responses have a greater proportion of 41BB-Lo CD8+ T cells compared to others. Two comparisons of interest include the similarity in the proportion of 41BB-Lo CD8+ T cells in stable vs. no ICB treatment (may suggest that stable response mirrors the untreated tumor) as well as the huge difference in the proportion between progressive and partial responses (may suggest overall differences in adaptive immunity responses to RCC). For the former, the Pearson's Chi-squared test provided a χ^2 of 20.24 and a p-value of 6.829e-06. This rejects the null hypothesis that there is no difference in the cell abundance of 41BB-Lo CD8+ T cells between stable and no ICB treatment. Against our initial considerations, the low p-value and high χ^2 suggests that a stable response can have a significantly higher proportion of 41BB-Lo CD8+ T cells among lymphoid cells. For the latter comparison, the Pearson's Chi-squared test provided a χ^2 of 448.19 and a p-value of 2.2e-16. This rejects the null hypothesis that there is no difference in the cell abundance of 41BB-Lo CD8+ T cells in progressive and partial response lymphoid cells. Aligning with our initial thoughts, the extremely low p-value and high χ^2 suggests that a partial response can have a significantly higher proportion of 41BB-Lo CD8+ T cells among lymphoid cells compared to a progressive response. Overall, these findings suggest how successful ICB treatment/response leads to a greater cell abundance of 41BB-Lo CD8+ T cells in lymphoid cells due to T cell exhaustion while the growing and metastasizing tumor experiences less of this.

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Figures and Tables

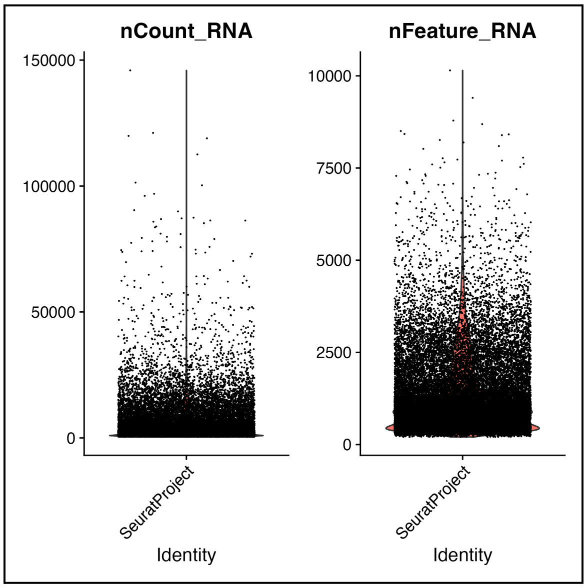

Figure 1: Violin plot of RNA Count and feature distribution in original dataset.

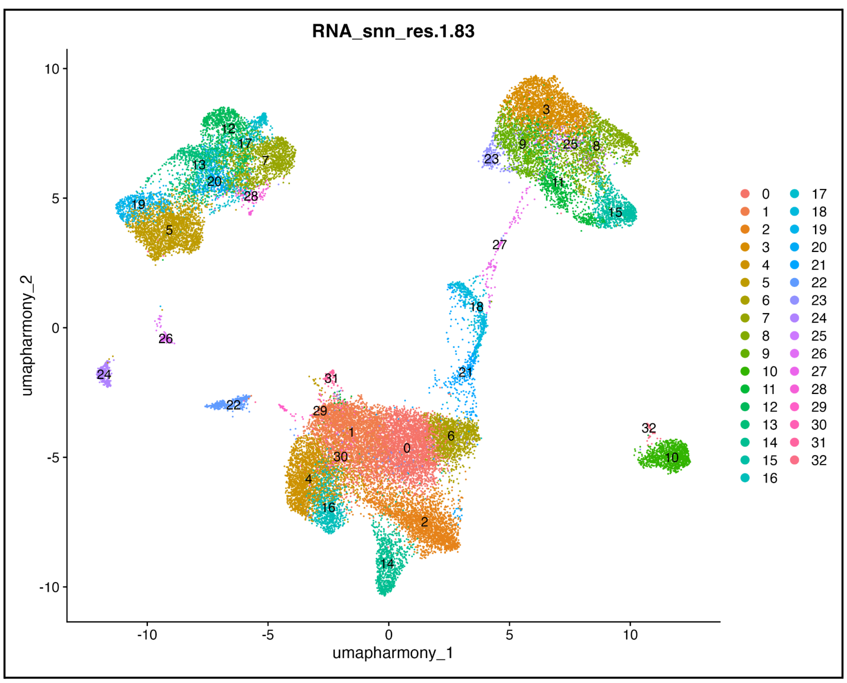

Figure 2: UMAP of 33 clusters from the entire dataset.

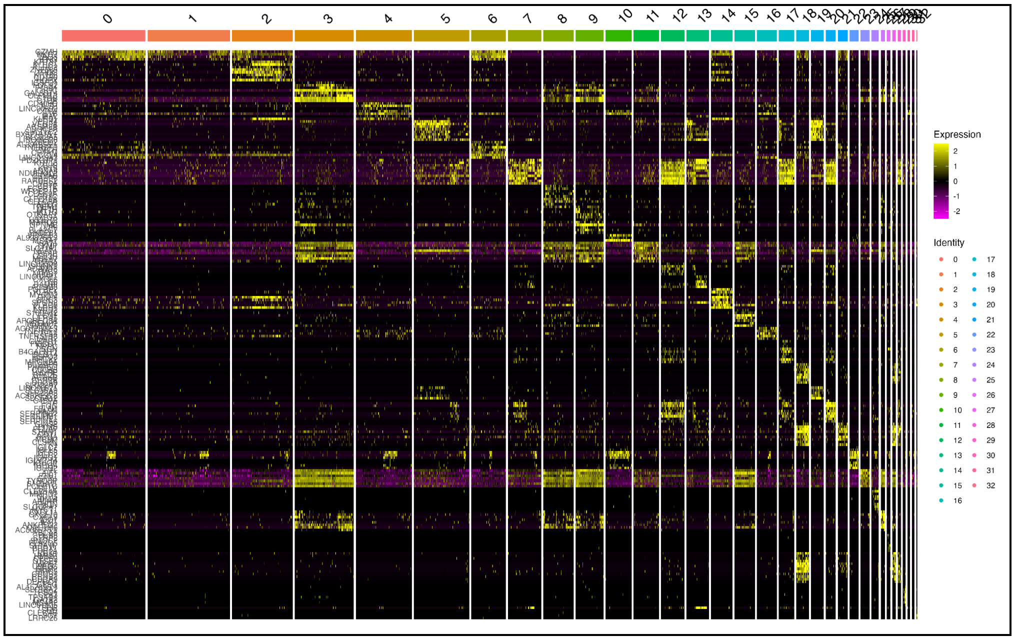

Figure 3: Heatmap of top marker genes for 33 clusters across the entire dataset.

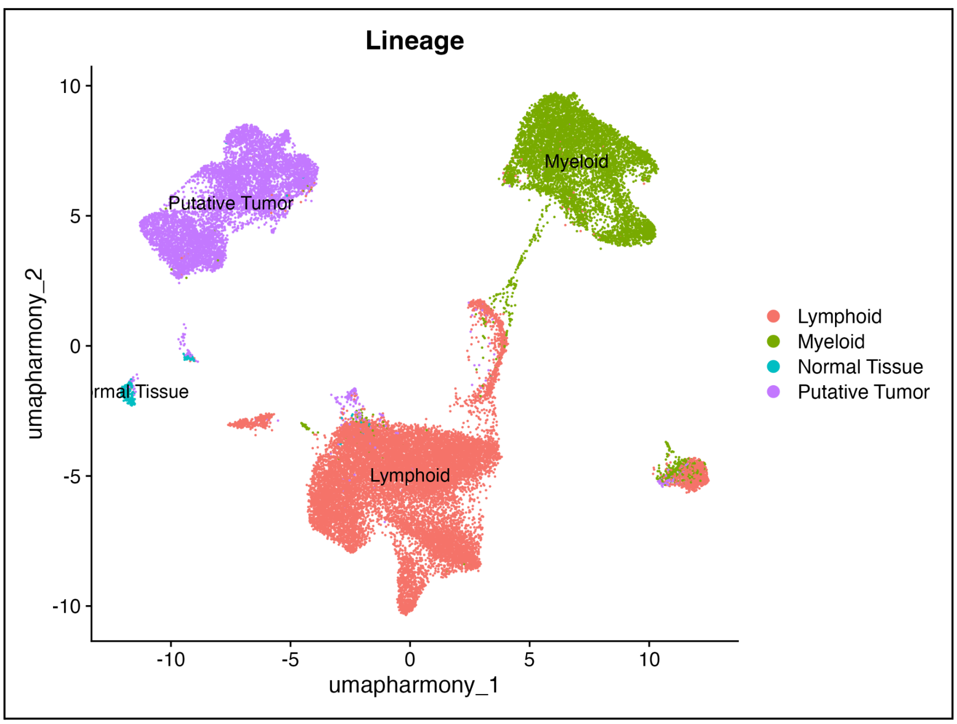

Figure 4: UMAP by cell types from the entire dataset.

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Table 1: Top marker genes for each lymphoid cell type cluster and identified cell type.

Cluster	Top Marker Genes	Identified Cell Type
0	CCL18, QPRT	Unknown
1	IL7R and CCR7, CD40LG and FOS	Memory and Effector T-Helper
2	CD8A, CD8B	41BB-Lo CD8+ T cell
3	CD8A, CD8B	41BB-Lo CD8+ T cell
4	HOPX, KLRD1	NKT
5	FOXP3, TNFRSF18	T-Reg
6	SPON2, S1PR5	FGFBP2+ NK
7	MS4A1, CD79A	B cell
8	MKI67, TOP2A	Cycling CD8+ T cell
9	XCL2, CCL1	41BB-Hi CD8+ T cell
10	KLRC1, GNLY	FGFBP2- NK
11	IGHA1, IGKC	Plasma cell
12	MKI67, TOP2A	Cycling CD8+ T cell
13	IFIT3, MX1	MX1-Hi CD8+ T cell
14	CBX3P4, GPR42	Unknown

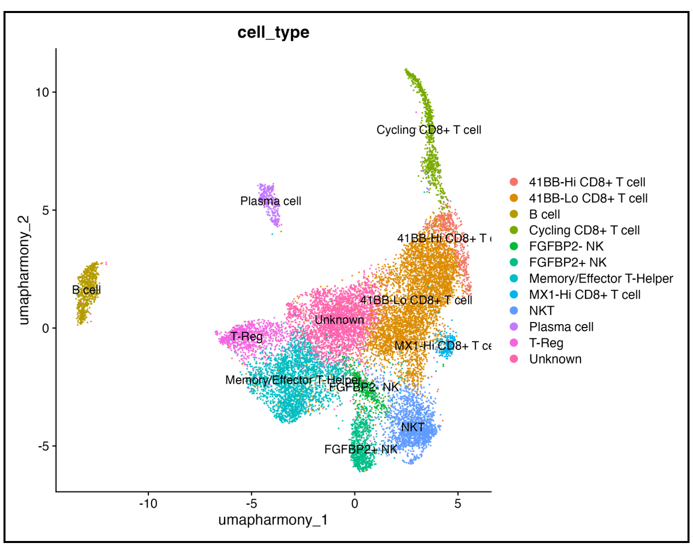

Figure 5: UMAP of lymphoid cell types.

Table 2: Top marker genes for each myeloid cell type cluster and identified cell type.

Cluster	Top Marker Genes	Identified Cell Type
0	GPNMB, CTSD	GPNMB-Hi TAM
1	CLEC10A, CD1C	CD1C+ DC
2	FCRL5, PAX5	Unknown
3	SELENOP, FOLR2, F13A1	FOLR2-Hi TAM
4	CXCL10, CXCL11, GBP1	CXCL10-Hi TAM
5	GPNMB, CTSD	GPNMB-Hi TAM
6	CD52, FCGR3A	CD16+ Monocyte
7	VCAN, S100A12, S100A8	CD16- Monocyte
8	PCLAF, STMN1	Cycling TAM
9	MIR3941, GPR22	Unknown
10	CD160, DTHD1	Unknown
11	CLEC9A, XCR1	CLEC9A+ DC
12	SCT, KCNA5	Unknown
13	GATA2, CTSG	Unknown

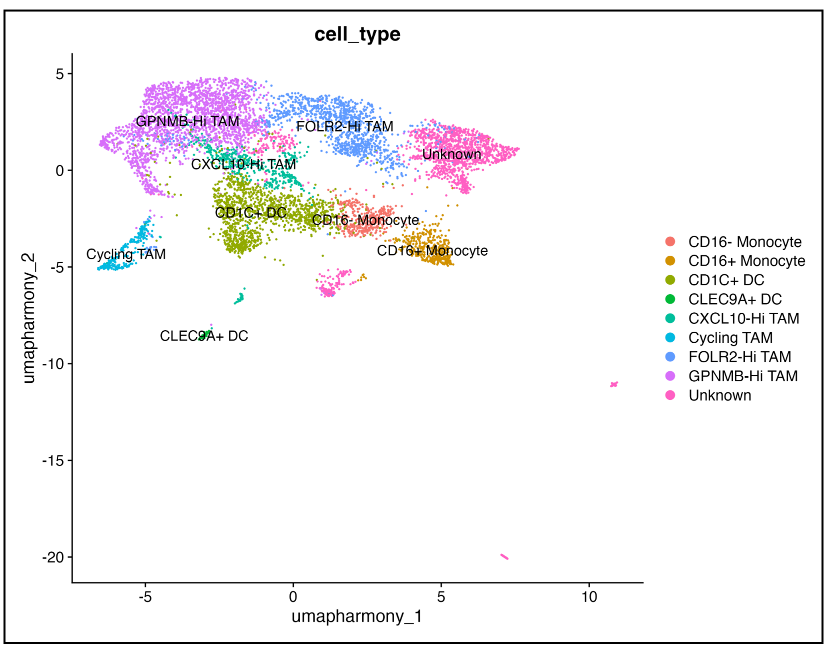

Figure 6: UMAP of myeloid cell types.

Table 3: Top three highly expressed pathways by ICB exposure in lymphoid cells.

ICB Treatment	Pathway #1	Pathway #2	Pathway #3
ICB	E2F Targets	G2M Checkpoint	MYC Targets V1
No ICB	Coagulation	Angiogenesis	Inflammatory Response

Table 4: Top three highly expressed pathways by ICB response in lymphoid cells.

ICB Response	Pathway #1	Pathway #2	Pathway #3
Progressive	Myogenesis	Epithelial Mesenchymal Transition	TGF Beta Signaling
Partial	E2F Targets	G2M Checkpoint	Interferon Alpha Response
Stable	TNFα Signaling via NFKB	Epithelial Mesenchymal Transition	Myogenesis
No ICB	Coagulation	Angiogenesis	Inflammatory Response
N/A	WNT Beta Catenin Signaling		

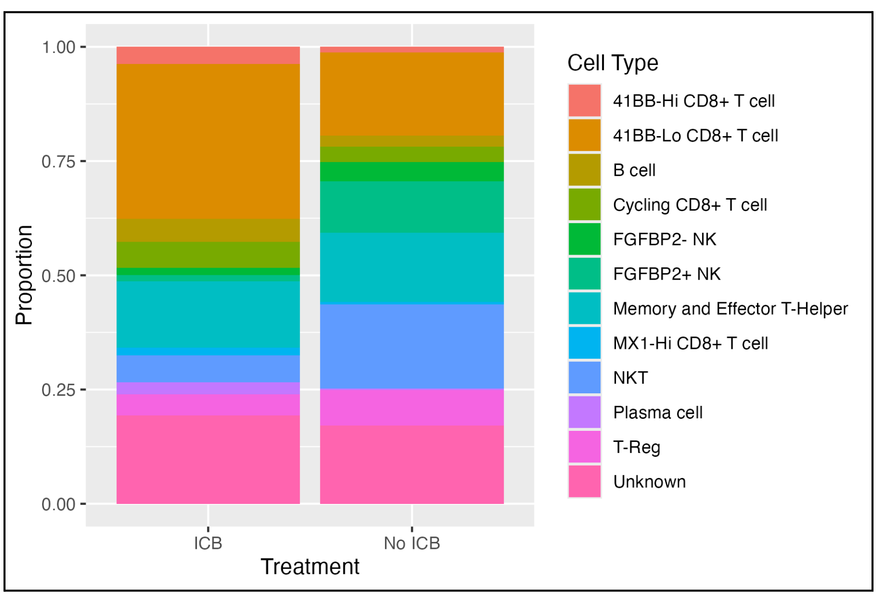

Figure 7: Proportion of each cell type by ICB exposure in lymphoid cells

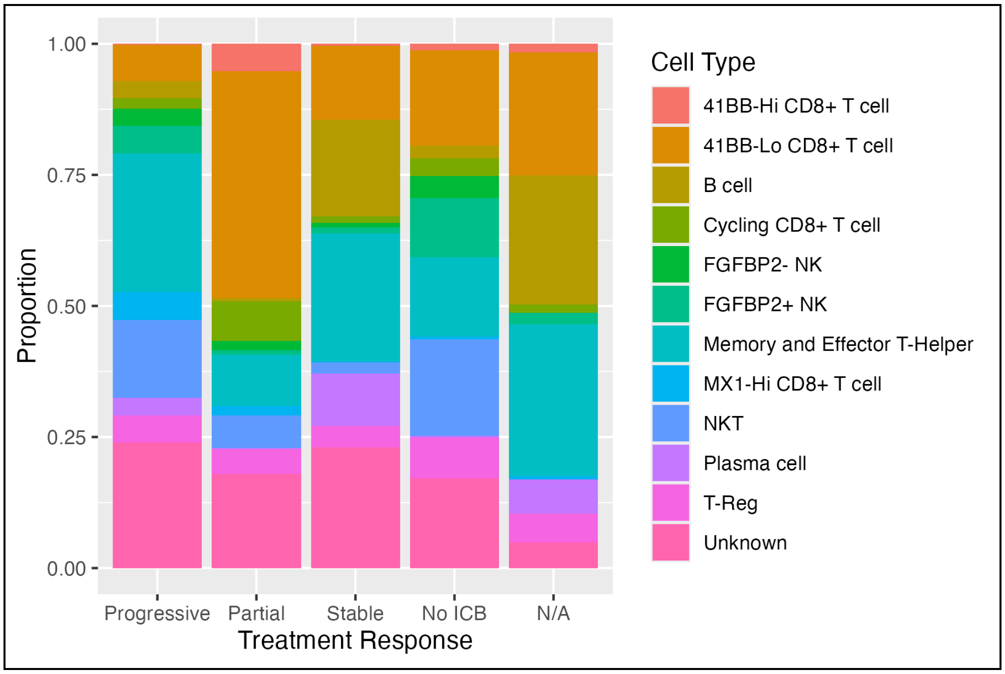

Figure 8: Proportion of each cell type by ICB response in lymphoid cells